

4.14

$$E(X) = 1000 = \frac{1}{\lambda} \rightarrow \lambda = 0.001$$

$$X \sim \text{Exp}(0.001)$$

a

$$\begin{aligned} P(X > 2000) &= \int_{2000}^{\infty} 0.001e^{-0.001x} dx \\ &= e^{-0.001 \cdot 2000} - 0 = e^{-2} \end{aligned}$$

b

$$\begin{aligned} P(X > 2000 | X > 500) &= \frac{P(X > 2000 \cap X > 500)}{P(X > 500)} = \frac{P(X > 2000)}{P(X > 500)} \\ P(X > 500) &= \int_{500}^{\infty} 0.001e^{-0.001x} dx \\ &= e^{-0.001 \cdot 500} - 0 = e^{-1/2} \\ P(X > 2000 | X > 500) &= \frac{e^{-2}}{e^{-1/2}} = e^{-3/2} \end{aligned}$$

4.49

Let X denote how long a unit lasts (in years). We know that $X \sim \text{Exp}(\frac{1}{10})$.

Our goal is to find (C, r) s.t. $0 = (C + 200) \times P(X > r) + (C - 600) \times P(X < r)$

$$\rightarrow (C + 200) \int_r^{\infty} \frac{1}{10} e^{-\frac{x}{10}} dx + (C - 600) \int_0^r \frac{1}{10} e^{-\frac{x}{10}} dx$$

$$\rightarrow (C + 200)(e^{-\frac{r}{10}}) + (C - 600)(1 - e^{-\frac{r}{10}})$$

$$\rightarrow C - 600 + 800e^{-\frac{r}{10}} = 0$$

$$C \text{ as a function of } r: C = 600 - 800e^{-\frac{r}{10}}$$

Thus, pairs $(C = 600 - 800e^{-\frac{r}{10}}, r = r)$ result in a profit of 0.

For $r = 2$, to result in 0 profit then $C = -54.98$ (I give the warranty holder ~\$50). Since I am expecting a profit of 200 dollars from people who do not buy the warranty I would price the warranty at a similar profit threshold and say the price for the full covered warranty is $C = 150$.

5.4

a

$$M_X(t) = e^{6t^2} \text{ when } |t| < 2$$

$X \sim N(0, 12)$ and bound by $|t| < 2$

b

$$M_Y(t) = \frac{2}{2-t} \text{ for } t < 0.5$$

$Y \sim \text{Exp}(2)$ and $t < 0.5$

c

$$M_Z(t) = \infty \text{ for } t \geq 5$$

Z could follow an exponential distribution with $\lambda \leq 5$

Thus we can determine Z as an exponential distribution but we cannot determine λ

d

$$M_W(2) = 2$$

$W \sim \text{Exp}(4)$ because for exponential distribution with $\lambda = 4$, $M(2) = \frac{4}{4-2} = 2$

Alternatively W could follow $P(W = 0) = 1/2$ and $P(W = \frac{\ln 3}{2}) = 1/2$

Thus $M_W(2) = \sum_i e^{2i} P(W = i) \rightarrow 1/2 e^{2 \cdot 0} + 1/2 e^{2/2 \ln 3} = 1/2 + 3/2 = 2$

Therefore not sufficient information to determine the distribution of W .

5.8

$$f_x = \frac{1}{b-a} \text{ if } x \in [a, b] \text{ and } 0 \text{ else}$$

$$f_x = \frac{1}{3} \text{ if } x \in [-1, 2] \text{ and } 0 \text{ else}$$

$$f_y(y) = \frac{f_x(g^{-1}(y))}{g'(g^{-1}(y))}$$

$$g(x) = x^2$$

$$g^{-1}(y) = \sqrt{y}$$

$$g'(x) = 2x$$

$$f_y(y) = \frac{1/3}{2\sqrt{y}} = \frac{1}{6\sqrt{y}}$$

Now the bounds of Y are not simply $-1^2, 2^2$ because this results in $[1, 4]$ which does not include values of $x \in (-1, 1)$. We can think about it as $x \in [-1, 1]$ maps to values of $y \in [0, 1]$, meaning this section is mapped twice and thus follows $\frac{1}{3\sqrt{y}}$

$$f_y(y) = \begin{cases} \frac{1}{3\sqrt{y}} & \text{if } x \in [0, 1) \\ \frac{1}{6\sqrt{y}} & \text{if } x \in [1, 4] \\ 0 & \text{else} \end{cases}$$

5.11

a

$$M_X(t) = \int_0^\infty e^{tx} x e^{-x} dx = \int_0^\infty x e^{x(t-1)} dx$$

Integration by parts:

$$\frac{x e^{x(t-1)}}{t-1} \Big|_0^\infty - \int_0^\infty \frac{e^{x(t-1)}}{t-1} dx$$

For $t < 1$ the first term goes to 0 so

$$\rightarrow 0 - \frac{e^{x(t-1)}}{(t-1)^2} \Big|_0^\infty$$

$$\rightarrow 0 + \frac{e^0}{(t-1)^2} \rightarrow \frac{1}{(t-1)^2}$$

$$M_X(t) = \frac{1}{(t-1)^2}, t < 1$$

b

$$E(X^n) = M_X^n(0) \rightarrow \frac{d^n}{dt} \left(\frac{1}{(t-1)^2} \right) \text{ then plug in 0 for } t$$

$$\rightarrow \frac{d^n}{dt} (t-1)^{-2}$$

$$\text{For } n = 1 \rightarrow \frac{-2}{(t-1)^{-3}}, t = 0 \rightarrow 2$$

$$\text{For } n = 2 \rightarrow \frac{6}{(t-1)^{-4}}, t = 0 \rightarrow 6$$

$$\text{For } n = 3 \rightarrow \frac{-24}{(t-1)^{-5}}, t = 0 \rightarrow 24$$

$$E(X^n) = (n+1)!$$

5.18

a

$$M_X(t) = E(e^{tX}) = \sum_{k=0}^{\infty} e^{tk} p(1-p)^{k-1}$$

$$\rightarrow \frac{p}{1-p} \sum_{k=0}^{\infty} e^{tk} (1-p)^k$$

$$\rightarrow \frac{p}{1-p} \sum_{k=0}^{\infty} (e^t(1-p))^k$$

To prevent the sum from diverging, $|e^t(1-p)| < 1 \rightarrow t < -\ln(1-p)$

$$\rightarrow \frac{p}{1-p} \frac{e^t(1-p)}{1-e^t(1-p)}$$

$$\rightarrow M_X(t) = \frac{pe^t}{1-e^t(1-p)}, t < -\ln(1-p)$$

b

$$E(X) = M'_X(0)$$

$$\rightarrow p \left(\frac{(1-e^t(1-p))e^t - e^t(0-e^t(1-p))}{(1-e^t(1-p))^2} \right)$$

$$\rightarrow p \left(\frac{(1-(1-p)) - (- (1-p))}{(1-(1-p))^2} \right)$$

$$\rightarrow p \left(\frac{1}{p^2} \right)$$

$$\rightarrow \frac{1}{p}$$

$$\text{Var}(X) = E(X^2) - E(X)^2 \rightarrow M''_X(0) - M'_X(0)^2$$

$$M''_X(0) = \frac{d}{dt} M'_X(t)$$

$$\begin{aligned}
&\rightarrow \frac{d}{dt} p \left(\frac{(1-e^t(1-p))e^t - e^t(0-e^t(1-p))}{(1-e^t(1-p))^2} \right) \\
&\rightarrow \frac{d}{dt} p \left(\frac{(1-e^t+pe^t)e^t - e^t(-e^t+pe^t)}{(1-e^t+pe^t)^2} \right) \\
&\rightarrow \frac{d}{dt} p \left(\frac{e^t}{(1-e^t+pe^t)^2} \right) \\
&\rightarrow p \frac{(1-e^t(1-p))^2 e^t - (e^t(2(1-e^t(1-p))(-e^t(1-p))))}{(1-e^t(1-p))^4} \\
&\rightarrow p \frac{(1-(1-p))^2 - (2(1-(1-p))(-1-p))}{p^4} \\
&\rightarrow \frac{(p^2) - (2p)(-1+p)}{p^3} \\
&\rightarrow \frac{(p) - (-2+2p)}{p^2} \\
&\rightarrow \frac{2-p}{p^2} \\
Var(X) &= \frac{2-p}{p^2} - \frac{1}{p}^2 = \frac{1-p}{p^2}
\end{aligned}$$

5.35

CDF of $Exp(\lambda) \rightarrow F(x) = 1 - e^{-\lambda x}, x \geq 0$

$$f_{\lfloor X \rfloor}(x) = F(x+1) - F(x), x \in \mathbb{N}$$

This represents the probability that a number lies in the range $[x, x+1)$ which is how the ceiling function works.

We can do a sanity check to see that our function works as a pmf.

$$\begin{aligned}
&\sum_{k=0}^{\infty} f_{\lfloor X \rfloor} \\
&\sum_{k=0}^{\infty} F(k+1) - F(k) \\
&-F(0) + F(1) - F(1) + F(2) - F(2) + F(3) - \dots + F(\infty) \\
&F(\infty) - F(0) \\
&1 - e^{-\infty} - (1 - e^0) = 1
\end{aligned}$$

6.3

a

$$(W, Y, Pu) \sim Mult(10, 3, 0.5, 0.4, 0.1)$$

$$P(W = 5, Y = 4, Pu = 1) = \binom{10}{5, 4, 1} 0.5^5 0.4^4 0.1$$

$$P(W = 5, Y = 4, Pu = 1) = \frac{10!}{5!4!1!} 0.5^5 0.4^4 0.1$$

$$P(W = 5, Y = 4, Pu = 1) = (2 * 3 * 4 * 7 * 6) * 0.5^5 0.4^4 0.1$$

$$P(W = 5, Y = 4, Pu = 1) = 0.08064$$

b

$$P(W = 9) = P(W = 9, Y = 1, Pu = 0) + P(W = 9, Y = 0, Pu = 1)$$

$$P(W = 9) = 4 * 0.5^9 + 0.5^9 = 5 * 0.5^9 = \frac{5}{512}$$